```
global_data = {

    # ── ATTRIBUTE NAME
    "attribute_1": "total_cost",
    "attribute_2": "material_continuity",
    "attribute_3": "conductor_cross_section_mm2",
    "attribute_4": "voltage_rating_kv",
    "attribute_5": "cable_length_km",

    # ── ATTRIBUTE DIRECTION
    "direction_attribute_1": "negative",
    "direction_attribute_2": "positive",
    "direction_attribute_3": "negative",
    "direction_attribute_4": "positive",
    "direction_attribute_5": "negative",

    # ── GLOBAL MIN / MAX BOUNDS ──────
    "attribute_1_min": 150000,       # USD/km/circuit
    "attribute_1_max": 3000000,

    "attribute_2_min": 0,            # binary — hard floor
    "attribute_2_max": 1,            # binary — hard ceiling

    "attribute_3_min": 95,           # mm²
    "attribute_3_max": 2500,

    "attribute_4_min": 69,           # kV
    "attribute_4_max": 500,

    "attribute_5_min": 1,            # km
    "attribute_5_max": 100,

    # ── COMMODITY PRICES ─────────────
    "copper_price_per_ton": 9500,
    "aluminum_price_per_ton": 2500,
  "copper_price_per_kg": 9.500,      # fixed global Cu price
    "aluminum_price_per_kg": 2.500,   # fixed global Al price
}


regional_data = [

    # ── UNITED STATES ───────────────────────────
```

```
# Cu-dominant market (~55% Cu); mixed but copper-preferring for urban HV underground.
# High voltage rating weight (0.20) reflects NERC/FERC tier-based procurement standards.
# Material continuity (0.20): moderate institutional inertia from incumbent IOU specs.
{
    "region": "United States",
    "copper_product_market_share": 0.55,   # Cu slightly dominant; aluminum growing
    "aluminum_product_market_share": 0.45,
    "weight_attribute_1": 0.30,  # total_cost
    "weight_attribute_2": 0.20,  # material_continuity
    "weight_attribute_3": 0.20,  # conductor_cross_section_mm2
    "weight_attribute_4": 0.20,  # voltage_rating_kv
    "weight_attribute_5": 0.10,  # cable_length_km
},

# ── CHINA ──────────────────────────────────────────
# Al-dominant market (~60% Al); State Grid cost-driven procurement at massive scale.
# Highest cost weight (0.40); low material continuity weight (0.10) — easy to switch.
# Cable length weight elevated (0.15): long national grid corridors amplify cost gap.
{
    "region": "China",
    "copper_product_market_share": 0.40,   # Al majority due to cost pressure
    "aluminum_product_market_share": 0.60,
    "weight_attribute_1": 0.40,  # total_cost
    "weight_attribute_2": 0.10,  # material_continuity
    "weight_attribute_3": 0.20,  # conductor_cross_section_mm2
    "weight_attribute_4": 0.15,  # voltage_rating_kv
    "weight_attribute_5": 0.15,  # cable_length_km
},

# ── INDIA ──────────────────────────────
# Strongly Al-dominant (~70% Al); extreme capex sensitivity, rapid grid expansion.
# Highest cost weight of all regions (0.45); material continuity minimal (0.10).
# Low cable_length weight (0.10): Indian underground projects tend to be shorter urban runs.
{
    "region": "India",
    "copper_product_market_share": 0.30,   # Al strongly dominant
    "aluminum_product_market_share": 0.70,
    "weight_attribute_1": 0.45,  # total_cost
    "weight_attribute_2": 0.10,  # material_continuity
    "weight_attribute_3": 0.20,  # conductor_cross_section_mm2
    "weight_attribute_4": 0.15,  # voltage_rating_kv
    "weight_attribute_5": 0.10,  # cable_length_km
},
```

```
    # ── EUROPE ──────────────────────────────────────
    # Strongly Cu-dominant (~75% Cu); premium specifications, reliability focus, deep Cu inertia.
    # Highest material_continuity weight (0.30): strongest institutional and historical lock-in.
    # Lower cost weight (0.25): total-cost-of-ownership logic over 40+ yr asset life reduces
    # raw price sensitivity; copper accepted at premium for compact duct bank fit.
    {
        "region": "Europe",
        "copper_product_market_share": 0.75,   # Cu strongly dominant
        "aluminum_product_market_share": 0.25,
        "weight_attribute_1": 0.25,  # total_cost
        "weight_attribute_2": 0.30,  # material_continuity
        "weight_attribute_3": 0.20,  # conductor_cross_section_mm2
        "weight_attribute_4": 0.15,  # voltage_rating_kv
        "weight_attribute_5": 0.10,  # cable_length_km
    },
]


product_data = [

    {
        "region": "United States",
        "dominant material": "cu",
        "product": "138kV XLPE Cu 500mm² Underground Transmission Cable",
        "copper_kg": 13440,        # kg per km of 3-phase circuit (conductor only)
        "aluminum_kg": 0,
        "total_mass_kg": 34440,     # full cable assembly per km circuit: (4480+7000)×3
        "non_material_cost_per_unit": 620000,   # USD/km/circuit (XLPE, screen
        "attribute_2_value": 1,     # US is Cu-dominant region → Cu matches prior material
        "attribute_3_value": 500,   # 500 mm² per-phase conductor cross-section
        "attribute_4_value": 138,   # 138 kV (ANSI standard US transmission)
        "attribute_5_value": 15,    # 15 km canonical urban corridor project
    },
    {
        "region": "United States",
        "dominant material": "al",
        "product": "138kV XLPE Al 800mm² Underground Transmission Cable",
        "copper_kg": 0,
        "aluminum_kg": 6480,        # kg per km of 3-phase circuit (conductor only)
        "total_mass_kg": 28980,     # (2160+7500)×3 — lighter conductor, slightly more insulation
        "non_material_cost_per_unit": 587000,   # USD/km/circuit
        "attribute_2_value": 0,     # US is Cu-dominant → Al breaks Cu history (no continuity)
        "attribute_3_value": 800,   # 800 mm² per phase (1.6× Cu for equiv. ampacity)
        "attribute_4_value": 138,
```

```
    "attribute_5_value": 15,
},


{
    "region": "China",
    "dominant material": "cu",
    "product": "220kV XLPE Cu 630mm² Underground Transmission Cable",
    "copper_kg": 16935,
    "aluminum_kg": 0,
    "total_mass_kg": 46935,     # (5645+10000)×3
    "non_material_cost_per_unit": 887000,
    "attribute_2_value": 0,     # China is Al-dominant → Cu breaks Al history
    "attribute_3_value": 630,
    "attribute_4_value": 220,   # 220 kV (GB standard China national grid)
    "attribute_5_value": 30,    # 30 km canonical inter-urban corridor project
},
{
    "region": "China",
    "dominant material": "al",
    "product": "220kV XLPE Al 1000mm² Underground Transmission Cable",
    "copper_kg": 0,
    "aluminum_kg": 8100,
    "total_mass_kg": 41100,     # (2700+11000)×3
    "non_material_cost_per_unit": 820000,
    "attribute_2_value": 1,     # China is Al-dominant → Al matches Al history
    "attribute_3_value": 1000,
    "attribute_4_value": 220,
    "attribute_5_value": 30,
},


{
    "region": "India",
    "dominant material": "cu",
    "product": "132kV XLPE Cu 300mm² Underground Transmission Cable",
    "copper_kg": 8064,
    "aluminum_kg": 0,
    "total_mass_kg": 26064,     # (2688+6000)×3
    "non_material_cost_per_unit": 430000,
    "attribute_2_value": 0,     # India is Al-dominant → Cu breaks Al history
    "attribute_3_value": 300,   # smallest Cu conductor in canonical set
    "attribute_4_value": 132,   # 132 kV (CEA/BIS standard India transmission)
    "attribute_5_value": 10,    # 10 km canonical urban smart-city project
```

```
    },
    {
        "region": "India",
        "dominant material": "al",
        "product": "132kV XLPE Al 500mm² Underground Transmission Cable",
        "copper_kg": 0,
        "aluminum_kg": 4050,
        "total_mass_kg": 23550,     # (1350+6500)×3
        "non_material_cost_per_unit": 426000,
        "attribute_2_value": 1,     # India is Al-dominant → Al matches Al history
        "attribute_3_value": 500,
        "attribute_4_value": 132,
        "attribute_5_value": 10,
    },


    {
        "region": "Europe",
        "dominant material": "cu",
        "product": "220kV XLPE Cu 800mm² Underground Transmission Cable",
        "copper_kg": 21504,
        "aluminum_kg": 0,
        "total_mass_kg": 51504,     # (7168+10000)×3
        "non_material_cost_per_unit": 1055000,
        "attribute_2_value": 1,     # Europe is Cu-dominant → Cu matches Cu history
        "attribute_3_value": 800,   # largest Cu conductor in canonical set
        "attribute_4_value": 220,   # 220 kV (IEC/CENELEC standard Europe)
        "attribute_5_value": 20,    # 20 km canonical urban undergrounding project
    },
    {
        "region": "Europe",
        "dominant material": "al",
        "product": "220kV XLPE Al 1200mm² Underground Transmission Cable",
        "copper_kg": 0,
        "aluminum_kg": 9720,
        "total_mass_kg": 44220,     # (3240+11500)×3
        "non_material_cost_per_unit": 965000,
        "attribute_2_value": 0,     # Europe is Cu-dominant → Al breaks Cu history
        "attribute_3_value": 1200,  # largest Al conductor in canonical set
        "attribute_4_value": 220,
        "attribute_5_value": 20,
    },
]
```